

Understanding Environmental Sustainability and Information Practices in Global South Fish Farming

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ABSTRACT

This project investigates environmental sustainability and information-sharing practices in the Global South, particularly among low-resource fish farming communities. Our ongoing ethnographic study with rural Bangladeshi fish farmers found that these communities develop an understanding of environmental sustainability through formal and informal training and via close-knit groups on social platforms. When communicating with other fish farmers on social platforms, they often confront misleading information about climate and environment and struggle to validate it. Therefore, the fish farmers form support groups to combat it collectively. We believe the findings of this work will inform HCI about how to grow mass awareness of environmental sustainability among farming professionals in low-resource areas using ICTs.

CCS CONCEPTS

• **Computer systems organization** → **Embedded systems**; *Redundancy*; **Robotics**; • **Networks** → *Network reliability*.

KEYWORDS

datasets, neural networks, gaze detection, text tagging

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1 INTRODUCTION

There is a growing interest in fish-based diets for better nutrition and for ethical reasons [22, 67, 81]. Therefore, the demand for fresh

fish is skyrocketing globally, along with massive agricultural interest in fish farming [5, 6, 57, 60, 76]. While fish-based meals have long been popular worldwide historically, fish farming is concentrated only within specific geographic areas as fish farming requires substantial environmental support [3, 4, 15, 32, 37, 59]. However, beneath the surface of this seemingly sustainable farming method lurks a complex web of environmental concerns.

The negative impacts of fish farming are often consequences of resource use and waste management strategies. For example, intensive aquaculture, particularly shrimp farming, often relies on groundwater extraction, leading to aquifer depletion and salinization of freshwater sources. This not only threatens drinking water supplies for local communities but also disrupts delicate coastal ecosystems [19, 24, 45]. Mangrove forests, vital for coastal protection and biodiversity, are frequently cleared for pond construction, disrupting habitats and exacerbating erosion [8, 12, 66]. Additionally, excess fish feed and excrement pollute water with nitrogen and phosphorus, triggering algal blooms that deplete oxygen and harm aquatic life [1, 36, 48, 53]. Reports have also shown that fish farmers often overuse antibiotics to combat diseases in densely packed ponds, leading to antibiotic resistance in fish and the surrounding environment [68, 73, 79]. The pesticides and herbicides used to control pests and weeds in the fish-farming water-bodies potentially contaminate the water and harm wildlife [10, 52]. These negative consequences of fish farming pose a serious threat to human health and ecosystem stability.

Research in environmental sustainability and agriculture has shown that the consequences of these environmental impacts are far-reaching, not only for the fish-farming areas (e.g., in Bangladesh) but also for the entire world [4, 38, 60, 61, 64, 77]. Reduced water quality, salinization, and erosion threaten public health, harms biodiversity, and decreases fish productivity in the long run. They jeopardize food security and livelihoods, particularly for vulnerable coastal communities. While these are few among many existing pressing concerns in environmental research, fish farmers' misconceptions and misinformation about environmental sustainability mean that they may not be able to adequately judge the risks and implications of their practices. As a result of these potentially massive externalities, global development initiatives and development sociology researchers are also increasing growing concerns in fish farming, related misinformation, and possible resultant risks. Even

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so, these remain understudied topics in human-computer interaction (HCI), sustainable-HCI, and social computing.

Our ongoing research fills this gap by investigating fish-farming practices in rural Bangladesh and their broader socio-technical context. Bangladesh, a Global South country in South Asia, has an intricate network of rivers, ponds, and coastal areas, making it a natural haven for fish farming. This centuries-old tradition is a crucial pillar of the nation's food security and economy, producing over 1.25 million tons of freshwater fish annually [4, 5, 60]. We have been engaging with more than 25 fish farmers in the southeastern part of the country and observed their farms for more than six months. Currently, we are conducting interviews and focus group discussions with them to understand their training processes and content, the different kinds of chemicals used in their profession, how they determine the possible resultant harms, and how they resolve their confusion and combat misconceptions in their farming. Our research is motivated by existing HCI4D literature on fish farming [18, 74]. Our research questions are:

RQ1: How are rural Bangladeshi fish farmers trained with modern farming practices, and to what extent do they rely on traditional and indigenous practices?

RQ2: What significant environmental sustainability concerns are prominent for farmers, which do they employ in practice, and which of them do they discard?

RQ3: How do the farmers manage encounters with suspicious information about their farming practice and the environment?

RQ4: How can different information and communication technologies help fish farmers in their profession and preserve the environment in the future?

While our project is ongoing, the findings so far make three contributions to HCI and sustainability in computing literature. First, we present how fish-farming practices are influenced by both formal training and local rural traditional knowledge, which include environmental concerns, and hence cause dilemmas among the low-literate farmers. Second, we discuss how rural fish farmers are influenced by misconceptions about the environment which are spread on social media and other ICTs. At the same time, farmers form support groups to combat such misconceptions collectively. Finally, we bring in fish farmers' insights into HCI about how to grow mass awareness of environmental sustainability among related farming professionals in low-resource areas using ICTs.

2 LITERATURE REVIEW

The history of sustainability scholarship in HCI is lengthy. For example, in early 2000 sustainability researchers in HCI addressed the renewal, repair, and reuse of tools and their material effects and connected them to broader ecosystem concerns [11, 41, 46, 78]. Researchers have also emphasized education, awareness campaigns across regions, and global collaborations in combating different environmental threats together [16]. A group of scholars have noted the value of the concept of persuasive sustainability that is married to individualism [7, 9, 14, 33, 34, 43, 47, 49, 50, 80], and urged stakeholders to move forward addressing complexities of social dynamics and interplay with the ecosystem as a collective

[17, 27, 30, 44, 58]. An additional body of literature has also emerged suggesting that individuals, especially conceived as consumers or users, do not have full control over their resource usage; instead, they are bound by social norms, economics, and existing infrastructure [17, 23, 25, 26, 75].

Most of today's environmental and sustainability research focuses on diverse issues, including humans and non-humans in the ecosystem, local culture and history, material practices, and more [51, 54–56]. Researchers have made an effort to counter the local effects of climate change and support more viable practices [21]. In this regard, HCI researchers have speculated employing crowdsourcing, geo-data, and streetmaps among others [29, 40, 62, 70]. Focus has also remained on designing affordable and reliable energy services for households and other daily activities [2, 20, 42]. In this regard, researchers have also emphasized designing and using solar energy-based tools and devices [72]. Research has also focused on crisis informatics and developed theories, frameworks, and tools [69, 70] which relate to sustainability. Another body of literature has also advanced sustainability and environmental research by emphasizing the concerns beyond humans [63, 65]. Agriculture and environment aspects have also remained a topic of interest in this scholarship [28, 31, 39, 56, 71]. All of the above-mentioned works have motivated us to expand HCI scholarship's understanding of environmental sustainability in the context of the Global South.

3 METHODS AND DATA

Our ethnographic study included semi-structured interviews, focus groups, and observations with fish farmers in three villages in Jessore, Bangladesh. Our access to participants in the villages was facilitated by the Rural Reconstruction Foundation (RRF), a non-profit global development organization that offers microfinance, education, health, and agricultural programs [35]. The NGO field-workers introduced us to their micro-credit clients who were fish farmers. After arriving in the village with the NGO workers, the ethnographer and first author held a public community meeting with the fish farmers and explained the purpose of the research study. We addressed their concerns and recruited participants from the meetings. They referred us to others in the community and thus further recruitment was performed through snowball sampling. We have so far interviewed and conducted focus groups with 25 fish farmers (8 female, 17 male; 18–55 years old). The first author is a native Bengali speaker born and raised in Jessore and has lived in various parts of Jessore. This positionality helped them to access the rural population and build rapport with participants. Under the supervision of an Institutional Review Board we obtained oral consent from the participants since many villagers were low-literate and would have trouble reading and understanding a written informed consent form.

3.1 Observation

We started our study by observing our participants' daily work, hangouts, regular meetings with other fish farmers, and collaborative activities, focusing on understanding how they gather and share knowledge of fish farming and how different environmental sustainability concerns are associated with their activities. We also asked them situated questions for the spontaneity and better

participation in this process. All the observation sessions and their note-taking were performed upon receiving permission from the participants.

3.2 Focus Group Discussion (FDG)

Discussion topics included farming training, local ecosystems, cultural practices, productivity, constraints, etc. We also investigated their current access to and use of technology (such as mobile phones, computers, the internet, etc.) and how they influenced local fish farming. We took detailed notes and audio-recorded all the discussions with the participants' permission.

3.3 One-on-one Interview

We conducted one-on-one semi-structured interviews with the participants at convenient venues, generally at their homes. Each interview lasted approximately thirty minutes. We asked them about their training in fish farming, the nature of tasks they perform in their profession, fish-farmer family dynamics, and the challenges they face in their work. Here, we investigated for further details and familial practices around climate and local environmental concerns. We took detailed notes of all the interviews and audio-recorded several of them with the participants' permission.

3.4 Data Collection and Analysis

We collected our data over the summer months in 2019 and 2022, based on the ethnographers' access to the field. The interruption in data collection happened due to the global COVID-19 situation. So far, we have 4 hours of audio recording, which we have translated and transcribed for thematic analysis [13]. While the data collection and analysis are still ongoing, we found several emerging themes are significant. In the following section, we present three such emerging themes.

4 FINDINGS

Jessore has evolved into a crucial center for fish farming, owing to its abundant water resources and favorable climatic conditions. Fish farming has become a substantial source of income for local communities, playing a pivotal role in alleviating poverty and more broadly in rural development. The success of fish farming in these regions has not only bolstered domestic food security but has also positioned Bangladesh as a global leader in fish and seafood exports. This industry has generated employment opportunities, particularly for women, who actively participate in fish processing and marketing. The government has also been instrumental in supporting the sector through initiatives such as training programs, subsidies, and infrastructure development, fostering the growth of aquaculture.

Through our fieldwork, we found that local fish farmers believed they had adopted sustainable and environmentally friendly practices. Many farmers have embraced integrated farming systems interlinking fish cultivation with rice paddies and vegetable crops. They told us this approach not only optimizes land utilization but also enhances nutrient cycling, leading to an overall improvement in productivity. Furthermore, integrating modern technologies, including advanced breeding techniques, water quality management, and efficient feed utilization, is believed to have heightened the

effectiveness of fish farming in Jessore. Below, we discuss local fish farming infrastructure, the challenges that fish farmers face, and how they involve different information and communication technologies in their practice.

4.1 Fish Farming infrastructure in Jessore

The primary species under cultivation in local fish farming areas encompass tilapia, pangasius, carp varieties, and shrimp. Therefore, the training on fish farming organized by government agencies, NGOs, private training institutes, and local fish farmer associations and networks mainly focus on these fish species.

4.1.1 Types of Training. We describe our findings under two major categories: formal and informal training.

Formal Training: We define **Formal Training** as the forms of training and mentorship that come from entities, such as Bangladeshi government-approved and subsidized project organizers, and at the end of which the trainees receive certification. Examples of such entities could be government institutions, non-government organizations (NGOs), and university-run research programs. Our participants informed us that the Department of Fisheries under the Ministry of Fisheries and Livestock, Bangladesh, offers training programs through its Jessore District Fisheries Office and Jessore Fish Seed Multiplication Farm, which are within 25 kilometers radius of the villages we conducted our ethnography in. These programs cover pond management, fish breeding, disease prevention, and marketing. In addition, several NGOs and related private institutes like Jessore Aquaculture Training Center, RRF Training And Research Center (TARC), and BRAC's Aquaculture Training Center offer short-term courses and workshops on specific aspects of fish farming, such as vaccination, antibiotic use, growth measurement, and pond disinfection. Also, Jessore Science and Technology University's Department of Aquaculture offers Bachelor's and Master's degrees in aquaculture, where local researchers often invite and train the fish farmers, conduct their research with them, and conduct awareness programs, too. In most of the cases these training sessions are paid. Our participants mentioned that formal training is beneficial for everyone, particularly for beginners to plan out their initial fish farms.

Informal Training: We define **informal training** as forms of training where fish farmers receive advice and learn techniques from their trusted sources, which may or may not be government authorized. This is performed at will by those training, and farmers do not receive any certification. Examples of such informal training could be peer learning, fish-farmer organizations' conversations, online tutorials, etc. Our participants informed us that experienced and well-off fish farmers in Jessore often share their knowledge and best practices through informal networks and mentorship programs, where they invite other farmers on occasion and host programs. They might also charge money for such mentorship. However, twelve of the fish farmers told us that their families were in the fish farming profession for generations and they learned the basics from the family members, neighbors, colleagues, and family friends. Also, local fish farmer unions frequently collaborate with many Government agencies and NGOs to organize field demonstrations showcasing new technologies and sustainable practices in fish farming. Our participants informed us that such informal

training is very timely and informative in most cases. They mentioned that informal training and knowledge sharing are helpful for farmers who require immediate suggestions and also for the poor fish farm-workers who may not have sufficient knowledge of fish farming from family or paid formal sources.

4.1.2 Focal Areas of Training: Our participants mentioned that the training they received from the above-mentioned sources is mostly on pond management, fish breeding and hatchery management, disease prevention and control, and market and business management. Seventeen participants mentioned that they received pond management training from BRAC and RRF, where they learned about sustainable pond management practices like water quality control, stocking density, feed management, and waste removal. Twenty of the participants told us that their training from the Ministry of Fisheries and Livestock's local branch in Jessore emphasized the techniques for breeding commercially important fish species like Rui, Catla, Mrigal, and Pangas are covered, along with hatchery operation and fry management. Another 15 participants mentioned that they regularly attend trainings and discussions on disease prevention and control organized formally and informally where they learn about common fish diseases, their symptoms, and preventive measures, including biosecurity protocols. All the participants mentioned that to survive in fish farming understanding of market and business management was crucial and they often turn to NGOs and local experts for their skills in fish marketing, value addition, and financial management to secure better profits and expand fish farming businesses.

4.2 Challenges in Fish Farming and Related Environmental Concerns

4.2.1 Unverified Benefits of Certain Chemicals and Additives. Our participants mentioned that they often faced challenges regarding antibiotic and hormone use, water treatment, and chemical use. For example, a fish farmer family of four showed us their farm and told us about their previous week's meeting with other neighboring fish farmers where they had a heated discussion on the growth-promoting effects of antibiotics and hormones. We quote one of the participants,

"Our neighboring pond owner used some unknown antibiotics to protect his Tilapia fishes from some kind of infection. He said he bought it from a salesman who gave him a good deal. While his fish's health did not make any progress, the huge rain and flood mixed up water and several of my Rui fishes died overnight. My greedy neighbor can not even say the name of the medicine and how much he used so that we can investigate more." (P5)

An additional 18 participants told us about false claims about the growth-promoting effects of **antibiotics and hormones** in fish feed and their concern about overuse, creating antibiotic resistance and hormonal imbalances in fish. This poses long-term health risks for consumers and potentially disrupts the ecosystem.

Twenty of the participants also told us that often the training organizers have their own agenda of fulfilling their reports or specific project goals and may end up putting excessive emphasis on some

specific part of the fish farming process over another. For example, a group of seven fish farmers attended a focus group discussion with us and discussed a newbie fish farmer's getting carried away by a trainer's misleading words and lost their fish. This group concluded that exaggerated claims about the efficacy of certain **water treatment chemicals** can encourage farmers to rely solely on them for pond management, neglecting crucial practices like water exchange and organic waste removal. This can lead to poor water quality and disease outbreaks.

4.2.2 Unrealistic Yield Expectations and Unproven Techniques: Our participants mentioned that most fish farmers in their area are low-income people, and therefore, they are required to borrow money from many sources to invest in the pond. However, the amount of money borrowed through loans from government agricultural banks and NGOs on relaxed terms may not suffice for many fish farmers; they end up borrowing from loan sharks in extreme conditions. Sometimes the loan sharks imposed high interest with an additional percentage on their final benefits. Note that most of our participants were low-literate, and they told us that they were not much skilled in their money management. Such situations burden the fish farmers with unrealistic fish production targets and put immense pressure on farmers to use unsustainable practices with higher short term yield. As a result, the farmers may start overstocking and use unhygienic fish food, leading to resource depletion and environmental damage.

For example, a group of our FGD participants informed us about a year-old rumor the local Tilapia fish market faced a plummet during COVID-19 in 2020. The rumor was about human waste from the hospitals being used as tilapia fish food for faster fish growth on local farms. Therefore, Tilapia sellers in the local market faced backlash from the community. They had to withdraw the fish from the market with a promise from the district administration of the local government that they would run proper investigations. We quote one of the participants,

"We know social media rumors are bad. It damages the people's life and also the society. But imagine this kind of rumor needs to break out for people to be alert. Suspicion about Tilapia farming was not new, but nobody took necessary actions or investigated it. So, this outburst was necessary. We do not know who is responsible for the rumor, but we actually thank them." (P11)

When we further discussed this rumor with another group of fish farmers in a second village. This group of seven participants told us that police harassed them because police would come and raid their fish food storage, randomly opening the food packages to verify what was inside. We quote one of them,

"Our village has had an excellent reputation in Tilapia farming for decades, which made us a target of the investigation. But who is going to tell them that we actually have better water and soil than many parts of the country, and our in-house-breed eggs (of fish) are high quality, so our fishes are larger and tastier than others? We do not need to feed our fishes any bad things. If you ask online people, they would not know, but if you



Figure 1: (a) a village fish hatchery where the fish-farmers grow fish eggs, larvae, and fry, and (b) a group of fish-farmers working in the pond, they were explaining to the ethnographer how they identify different health concerns of their fish.

ask fish farming trainers and the University's people, they do their research, they would know." (P3)

However, all the participants agreed that some farmers get swayed away by greed and adopt unproven "Miracle" Techniques for quick profits. The participants opined that such hype surrounding unproven or untested aquaculture practices can entice farmers to adopt them without proper evaluation of risks and effectiveness and can result in financial losses and potentially harm fish health and pond ecosystems.

4.2.3 Misconception about Environmental Impacts. Our participants mentioned that they would ask fellow fish farmers in the neighborhood whenever they encountered any suspicious information or technique from hearsay or on social media. They said they would also save such social media posts to bring those to the discussion table at their local fish farmers' social gatherings that happen once or twice every week. Twelve participants mentioned that in such discussions, generally, other knowledgeable farmers break down the information concerning the community, seek everyone's opinion and knowledge from experiences, and help them analyze it collectively.

However, some misconceptions, ignorance, and the resultant impact that the participants often experienced were beyond the scope of them taking collective notes. To further explain such a situation, our participants informed us that local fish farming and agricultural training are coordinated separately, which brought disadvantages to several of their grain and fish farms. More than 22 of the participants told us they had their fish farms and paddy fields side-by-side, and when heavy rain happened during the monsoon, the pesticides and chemicals washed up and mixed with each other's water. Most of the fish farmers did not know which of the ingredients they had used for fish farming could harm the paddy, and neither did the paddy farmers know which of their pesticides and solid food were bad for the fish. We quote from one of our participants,

"My pond submerged and flooded the neighbor's paddy field. And the waters mixed up. My fish even went to their field for a swim. Once the water came down the following week, almost 25 kgs of my fish died overnight. And their field produced trashy rice that year. Later,

when we discussed this with the NGO trainer, we found out that he was using fertilizers that were good for paddy but bad for fish, while my fish was eating food that shouldn't have been given to paddy. My neighbor and I did not know; nobody told us. We both were damaged."(P19)

Therefore, this group and many other participants urged for ways for all the parties involved with the environment, water, and soil to be well-informed about the possible damages caused by their choice of ingredients used in their work so that negligence may not be a reason for ecological degradation and hence, everyone's damage.

4.2.4 Spread of Information through Informal Channels. Our participants informed us that they frequently seek peer support in person and through online sources. Fish farmers in all three villages mentioned having dedicated Whatsapp groups to coordinate with other fish farmers. Fifteen participants told us they subscribed to many Bangladeshi and international fish farming YouTube channels to learn about new techniques. All the fish farmers told us they had joined and followed the Facebook group run by the Ministry of Fisheries and Livestock, Bangladesh.

However, many fish farmers shared their concerns that relying on informal channels like word-of-mouth and local market sources for information can expose the farmers to inaccurate or outdated practices. They also mentioned that the record shows that Bangladesh currently has more than three million fish farmers, while only 40,000 have access to the Ministry's Facebook page. The rest of the fish farmers might be beyond the immediate communication with the ministry and, highly likely, beyond other trusted sources. Our participants expressed the vulnerability of such fish farmer groups to misinformation.

5 DISCUSSION AND FUTURE DIRECTION

Findings from this ongoing research have brought our attention to several themes that cohere with emerging concerns in the intersection of sustainability, policy making, and HCI. **First**, our participants mentioned that local agriculture and fish farming sectors

are not coordinated by the government, while from their experiences, they could perceive that both sectors seek benefits and may cause damage to the ecosystem. **Second**, our participants shared their concerns about unverified information from informal channels and their potential harms. This set of concerns brought our attention back to designing transparent and reliable information platforms and finding better strategies to combat misinformation. While SIGCHI researchers have made substantial progress in designing such tools for Western societies, Global South has remained significantly underrepresented in that body of literature. **Third**, most of our participants were low-literate and had poor skills in their money management, thus taking loans from multiple sources and managing milestones with their timelines were complicated for the participants, causing them extreme burden and stress. This concern brings our attention back to designing appropriate FinTech tools and frameworks for such low-resource and low-literate communities. Our research needs to be taken to an advanced stage for more possible design implications. Also, note that our findings may vary from other parts of the Global South and we are not looking to generalize our findings.

In future research, we will find opportunities to help fish farmer communities with easier access to information and communication technologies by designing more appropriate tools for them. We plan to collaborate with other stakeholders such as the local NGOs and their trainers, the local government representative working in this sector, and fish farmers' unions. Our goal is to provide a holistic perspective on this sector so that researchers working in this domain may contribute with better designs of ICTs and help these low-resource communities in their profession.

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